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**Design and Implementation of a
(BLDC) Motor Controller**

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1. Introduction

This project presents the design and implementation of a Brushless DC (BLDC) Motor Controller, also known as an Electronic Speed Controller (ESC). The main objective of the project is to control the speed and switching sequence of a three-phase BLDC motor using a MOSFET-based inverter and dedicated gate driver circuits.

The controller was designed for mechatronics and robotics applications that require high efficiency, compact size, and reliable motor control. The system includes power electronics, gate driver circuitry, PWM control, protection methods, PCB implementation, and simulation using Simulink.

The project specifications required the following: [filecite?turn0file0?](#)

- Three-phase inverter using six N-channel MOSFETs.
- High-side and low-side gate drivers with bootstrap circuitry.
- PWM signal interface.
- Current sensing and protection.
- Simulink simulation.
- Technical documentation and hardware prototype.

2. Project Objectives

The main objectives of the project are:

1. Design a three-phase BLDC inverter using MOSFETs.
2. Implement proper gate driving for both high-side and low-side switches.
3. Generate PWM switching signals.
4. Ensure safe operation using dead-time techniques.
5. Build a hardware prototype and PCB.
6. Simulate the complete system using MATLAB Simulink.
7. Analyze and solve practical hardware problems during implementation.

3. System Overview

The BLDC motor controller consists of the following main sections:

1. Power Supply Section.
2. Gate Driver Section.
3. MOSFET Three-Phase Inverter.
4. PWM Control Signals.
5. Protection and Bootstrap Circuit.
6. BLDC Motor Load.

The controller converts DC power from the battery into controlled three-phase AC switching signals that drive the BLDC motor.

4. Hardware Components

4.1 MOSFET Bridge

The inverter stage uses six N-channel MOSFETs arranged in a three-phase bridge configuration.

Configuration

- 3 High-side MOSFETs.
- 3 Low-side MOSFETs.
- Each motor phase is connected between one high-side and one low-side switch.

Function

The MOSFET bridge performs electronic commutation by switching the phases in sequence according to the PWM signals.

Advantages of N-Channel MOSFETs

- Lower $R_{ds(on)}$.
- Higher efficiency.
- Better thermal performance.
- Faster switching speed.

4.2 Gate Driver Circuit

The project uses the IR2110 high-side/low-side gate driver IC.

Purpose of the Gate Driver

The IR2110 is used because the microcontroller cannot directly drive the MOSFET gates at the required voltage and current.

Driver Pins

- VCC: Driver power supply.
- COM: Ground reference.
- HIN: High-side input.
- LIN: Low-side input.
- HO: High-side gate output.
- LO: Low-side gate output.
- VB: Bootstrap supply.
- VS: High-side floating reference.

Driver Operation

- LO drives the low-side MOSFET directly.
- HO drives the high-side MOSFET using the bootstrap circuit.

4.3 Bootstrap Circuit

The bootstrap circuit is necessary for high-side MOSFET operation.

Components

- Bootstrap diode.
- Bootstrap capacitor.

Operation Principle

When the low-side MOSFET turns ON, the bootstrap capacitor charges from VCC through the diode. When the high-side MOSFET is required to switch ON, the stored voltage inside the capacitor is used to raise the gate voltage above the source voltage.

Important Relation

The gate driver requires:

$$V_B - V_S \approx 10V \text{ to } 12V$$

for proper high-side switching.

4.4 PWM Control

PWM signals are applied to the HIN and LIN pins of the driver.

PWM Function

- Controls motor speed.
- Controls switching timing.
- Reduces power loss.

Dead-Time

Dead-time is inserted between high-side and low-side switching to prevent shoot-through.

Shoot-through occurs when both MOSFETs in the same phase conduct simultaneously, causing a short circuit across the power supply.

5. Circuit Description

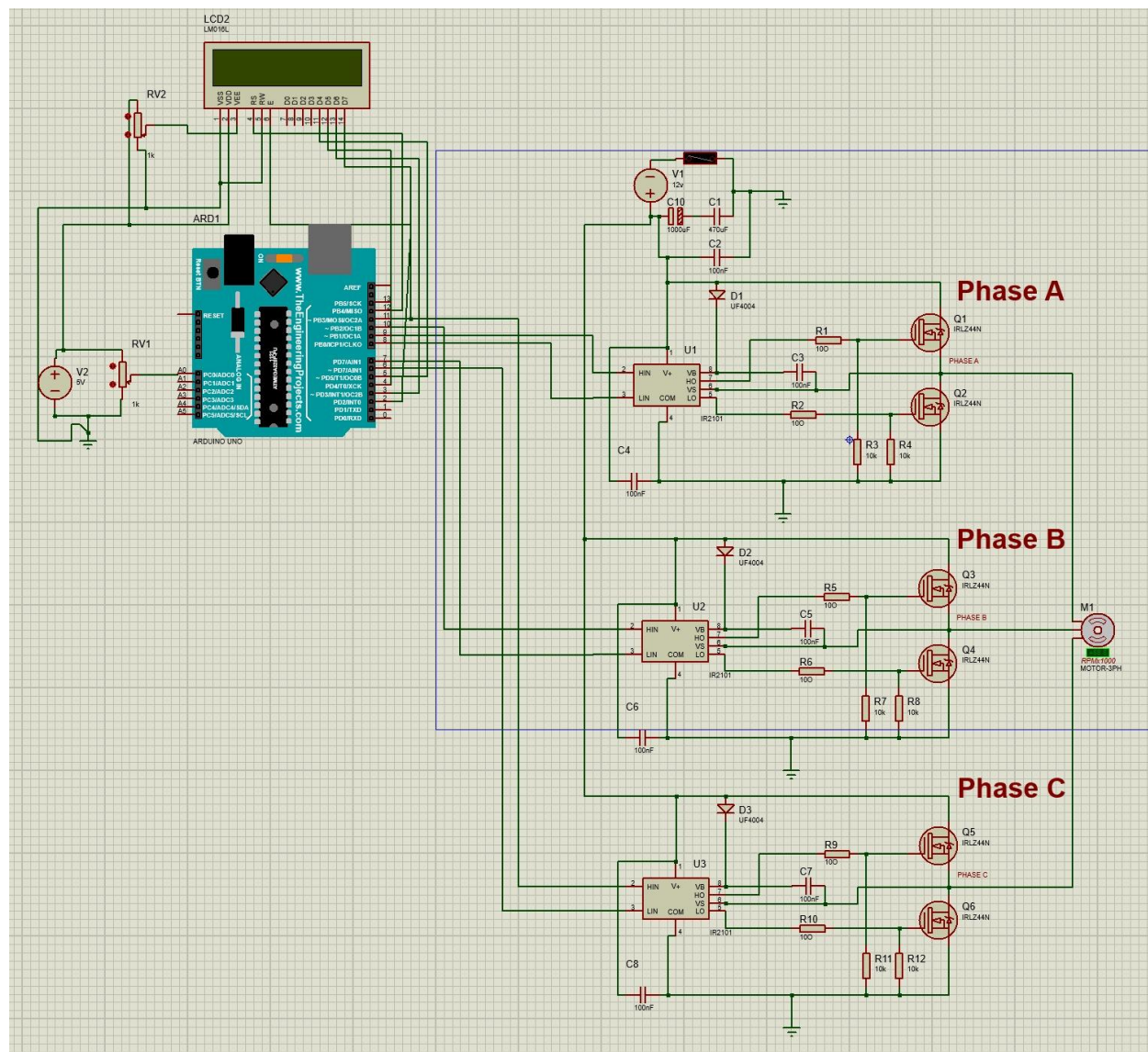
The complete circuit consists of three identical half-bridge stages.

Each stage contains:

- One high-side MOSFET.
- One low-side MOSFET.
- One IR2110 driver connection.
- One bootstrap capacitor.
- One bootstrap diode.

The outputs of the three bridge stages are connected to the three phases of the BLDC motor.

The DC supply powers the inverter, while the PWM signals control the switching sequence.



6. Simulink Simulation

A full simulation model was implemented using MATLAB Simulink.

Simulation Blocks

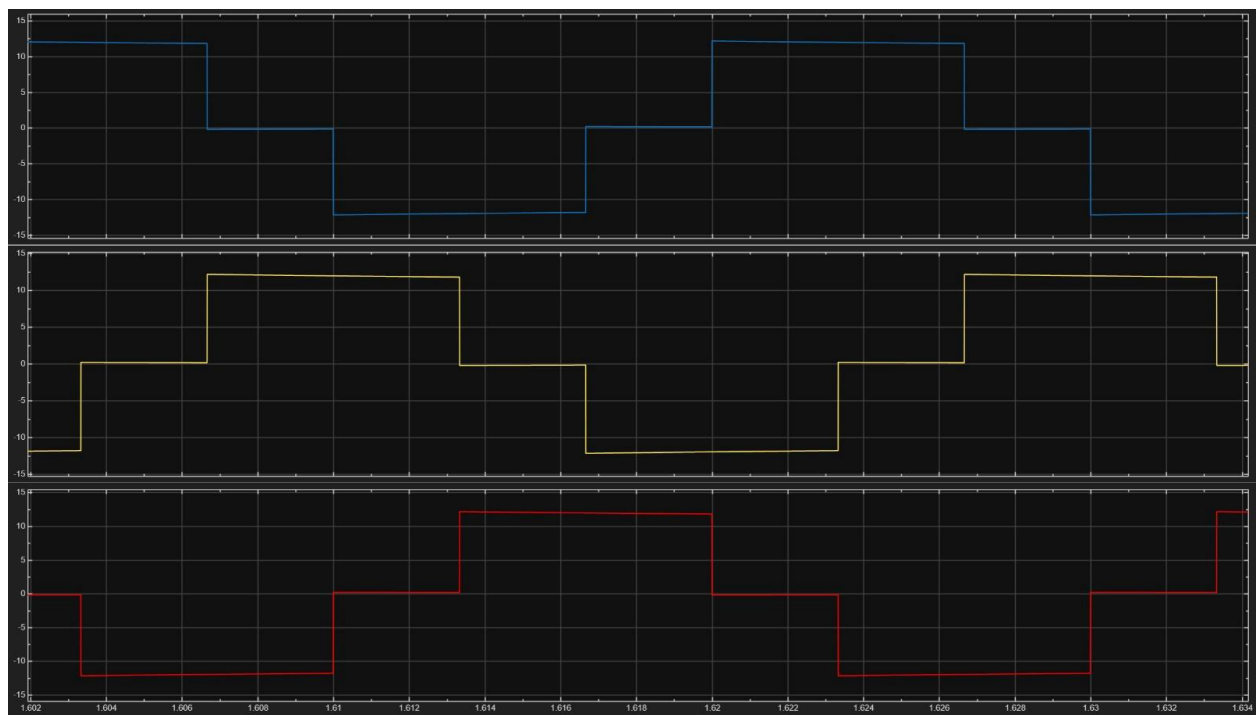
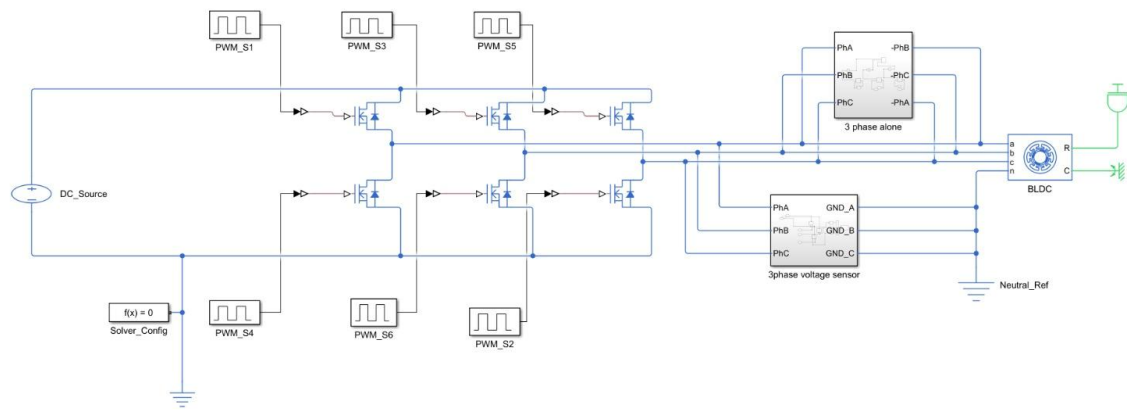
The simulation includes:

- DC Voltage Source.
- Three-phase inverter.
- PWM generation.
- Gate driver behavior.
- BLDC motor model.
- Measurement blocks.

Simulation Objectives

The simulation was used to:

- Verify switching operation.
- Check phase voltages.
- Analyze motor response.
- Detect short circuits.
- Validate PWM timing.



Simulation Problems Encountered

Problem 1: Motor Not Rotating

Initially, the motor did not rotate because the three motor phases were accidentally connected together in the Simulink model through an incorrect junction.

Solution

The phase connections were separated correctly so that each inverter output was connected independently to the motor phase.

Problem 2: Incorrect Grounding

The simulation experienced issues due to incorrect common grounding between voltage sources.

Solution

The grounds were unified correctly while maintaining separate phase outputs.

7. Prototype Implementation Using Perforated PCB

The hardware prototype was implemented using a perforated PCB board instead of a professionally manufactured printed PCB.

This implementation method was selected because it offers:

- Fast prototyping
- Easy hardware modification
- Lower manufacturing cost
- Faster debugging and troubleshooting

All electronic components were soldered manually on the perforated board using point-to-point wiring connections.

Implementation Considerations

Several practical considerations were taken into account during the hardware assembly process:

- Keeping power connections as short as possible
- Organizing wiring to reduce electrical noise
- Separating control signals from power lines
- Ensuring proper grounding between all circuit sections
- Reducing loose wire connections to improve reliability

Advantages of Using Perforated PCB

- Easy component replacement
- Flexible circuit modification
- Suitable for educational and prototyping purposes
- Faster implementation during testing stages

Although the perforated PCB does not provide the same performance as a professionally manufactured PCB, it was sufficient for testing, debugging, and validating the BLDC motor controller operation.

8. Hardware Testing

After PCB assembly, the circuit was tested step-by-step.

Initial Power Checks

The following voltages were verified:

- V_{CC} to $COM = 12V$
- V_{DD} to $V_{SS} = 5V$

The low-side outputs operated correctly, but issues appeared in the high-side operation.

9. Problems Encountered During Hardware Implementation

A major part of the project involved troubleshooting and debugging practical hardware issues.

Problem 1: High-Side MOSFET Not Switching

Observation

- LO output operated normally.
- HO output remained inactive.
- $V_B - V_S$ measured approximately 0V.

Analysis

The high-side driver requires a charged bootstrap capacitor. Since $V_B - V_S$ was zero, the driver could not generate the high-side gate voltage.

Root Cause

The bootstrap capacitor was connected in reverse polarity.

Solution

The capacitor orientation was corrected according to the polarity requirements between V_B and V_S .

After correcting the capacitor direction, the bootstrap voltage operated correctly.

Problem 2: Only One Phase Working Correctly

Observation

One phase responded correctly to HIN and LIN signals while the other phases showed only low-side switching.

Analysis

This indicated that the high-side bootstrap section was not operating correctly on the affected phases.

Solution

The following were checked:

- Bootstrap capacitor polarity.
- Bootstrap diode direction.
- MOSFET connections.
- VS node connection.
- Soldering quality.

The issue was resolved after correcting the bootstrap wiring and verifying all solder joints.

Problem 3: Incorrect MOSFET Driver Connections

Observation

The driver outputs did not respond correctly despite valid PWM signals.

Analysis

Some pins were incorrectly connected during soldering.

Solution

The driver pin connections were reviewed carefully according to the datasheet and rewired correctly.

Problem 4: Shoot-Through Risk

Observation

There was a risk of enabling high-side and low-side MOSFETs simultaneously.

Solution

Dead-time was inserted between switching transitions to protect the MOSFET bridge.

10. Safety Features

10.1 Dead-Time Insertion

Dead-time prevents simultaneous conduction between the high-side and low-side MOSFETs.

Benefits:

- Protects MOSFETs.
 - Prevents short circuits.
 - Improves reliability.
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10.2 Over-Current Protection

A shunt resistor can be used to monitor motor current.

If excessive current is detected:

- PWM signals can be disabled.
 - The inverter can be shut down.
-

10.3 Thermal Protection

The system design considers thermal dissipation using:

- Wide traces.
- Heat spreading copper areas.
- Proper component spacing.

11. Working Principle of the BLDC Controller

The BLDC controller operates by sequentially energizing the motor phases.

At any instant:

- One phase is connected to positive supply.
- One phase is connected to ground.
- One phase remains floating.

The switching sequence creates a rotating magnetic field that causes the rotor to rotate.

PWM is used to control the effective voltage applied to the motor, which directly controls speed.

12. Advantages of the Designed Controller

The designed ESC provides:

- High efficiency.
- Compact size.
- Fast switching.
- High power density.
- Reliable gate driving.
- Expandability for future control algorithms.

13. Future Improvements

Future enhancements may include:

1. Closed-loop speed control.
2. Hall sensor feedback.
3. Sensorless commutation techniques.
4. Current limiting and protection algorithms.
5. Regenerative braking functionality.
6. Wireless monitoring and communication.
7. LCD or OLED monitoring interface.
8. Field-Oriented Control (FOC) implementation for smoother and more efficient motor operation.

14. Conclusion

This project successfully demonstrated the design and implementation of a **Brushless DC (BLDC) Motor Controller** using a three-phase MOSFET inverter and IR2110 gate drivers.

The project combined several important engineering fields including:

- Power electronics
- Motor control systems
- PWM switching techniques
- Hardware implementation
- PCB prototyping
- Simulation and testing

Throughout the implementation process, several practical hardware and simulation problems were encountered, especially in the bootstrap and high-side driver sections. These issues were analyzed carefully and solved systematically through troubleshooting, measurements, and circuit verification.

The project provided valuable practical experience in:

- MOSFET switching and gate driving
- PWM generation and dead-time insertion
- Simulink simulation and verification
- Hardware debugging and testing
- BLDC motor operation principles

The final controller successfully achieved the required objectives and represents an important step toward more advanced motor control systems used in robotics, UAVs, electric vehicles, and industrial automation applications.

15. References

1. IR2110 Datasheet.
2. MATLAB Simulink Documentation.
3. MOSFET Datasheets.
4. BLDC Motor Control Theory.
5. Brushless DC (BLDC) Motor Controller Design Specifications.